Lab 06 – GUI

Lab Outcomes:

- Understand the Basics of Windows Forms: Learn the fundamentals of creating graphical user interfaces (GUI) in Windows Forms applications within the .NET framework.
- Design User Interfaces with Controls: Gain experience in using basic controls like buttons, labels, text boxes, checkboxes, radio buttons, and combo boxes to create interactive forms.

GUI

A Graphical User Interface (GUI) is a visual interface that allows users to interact with software or hardware through graphical elements like windows, buttons, icons, and menus rather than text commands or code. GUIs are designed to be intuitive and user-friendly, helping people easily navigate applications and perform tasks without needing specialized programming knowledge.

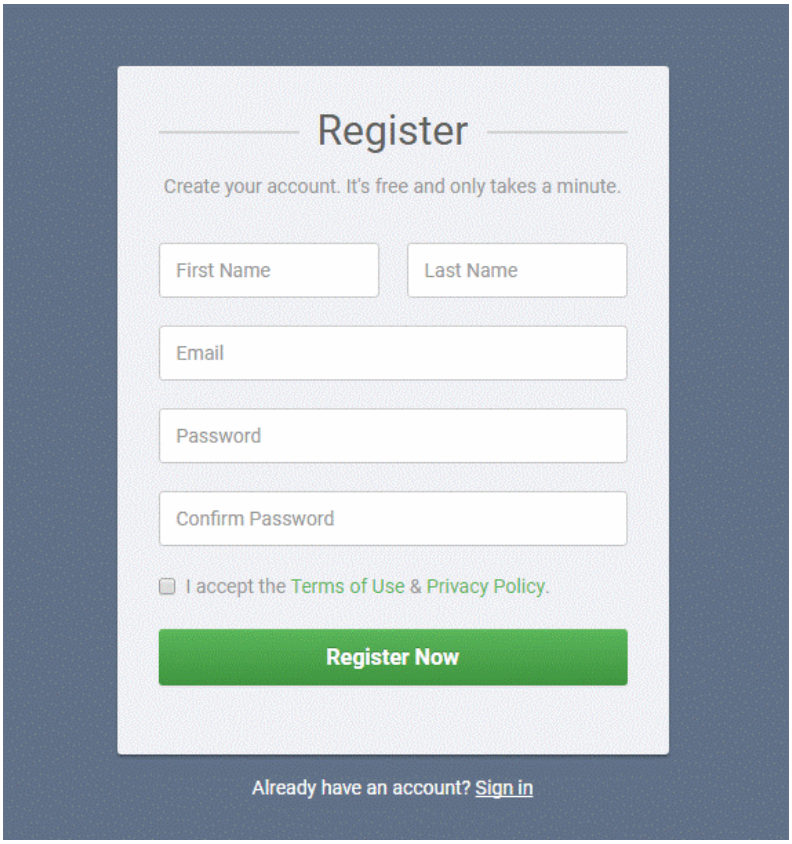
Key Elements of GUI

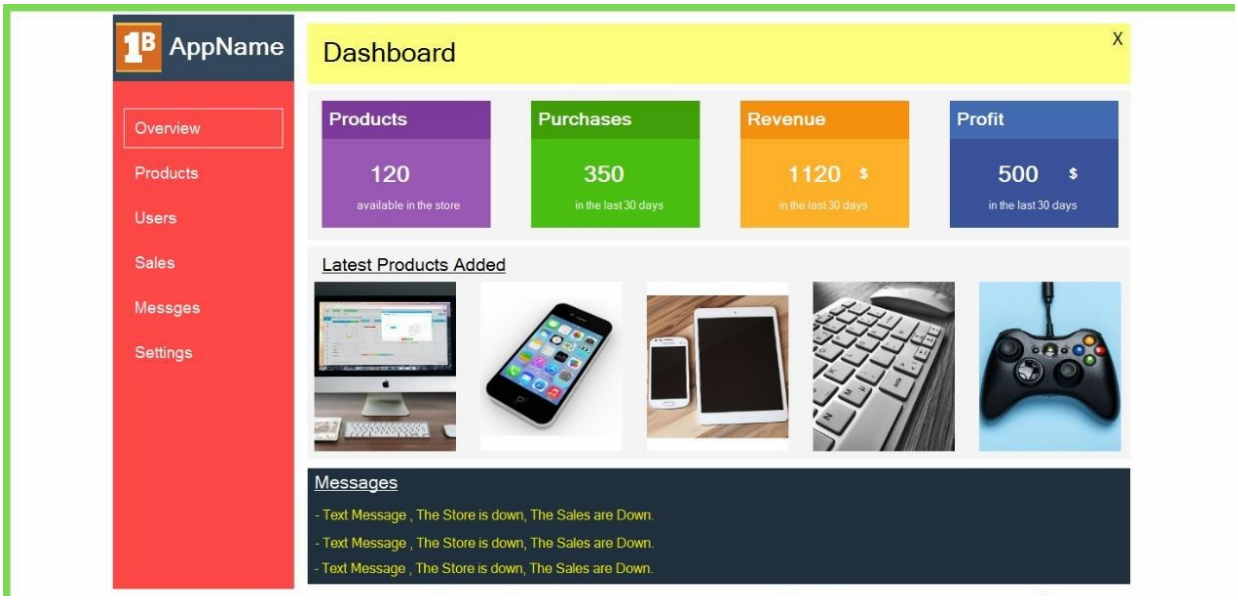
1. Windows and Forms: The main container for application content. Windows and forms serve as the framework where controls and elements are organized.
2. Controls: Interactive elements like buttons, text boxes, drop-down lists, checkboxes, and sliders. Each control performs specific functions, like allowing data entry or triggering actions when clicked.
3. Icons and Images: Visual representations of applications, files, or functions that help users recognize and access specific features quickly.
4. Menus and Toolbars: Allow users to access different parts of an application. Menus organize options hierarchically, while toolbars provide quick access to frequently used commands.
5. Navigation Components: Items like tabs, navigation bars, and links that allow users to move through different parts of the application.
6. Feedback and Dialogs: Includes progress bars, error messages, and pop-up dialogs that inform users about actions, warnings, or errors and guide them in correcting issues.

GUI in .NET Windows Forms

In the .NET framework, **Windows Forms** is a popular library for creating GUIs. It provides a range of prebuilt controls and tools that developers can use to build interactive applications without needing to code the layout from scratch. Windows Forms applications are well-suited for desktop applications that require rich interaction, data entry, and visualization tools.

Tasks: Create GUI Window Form in C# of the following given figures.





Lab Assessment Criteria:
Lab #06 Marks distribution

	ER1	ER6	ER8
Task	3 points	3 points	4 points

Lab # 06
Rubric Evaluation Guideline:

#	**Qualities & Criteria**	**0 < Poor <=1**	**1 < Satisfactory <= 2**	**2 < Excellent <=3**
ER1	Task/Program Completion	Minimal or no task functionality was achieved	Some tasks were completed but few has errors or incomplete functionalities.	All tasks were completed without any errors.
#	**Qualities & Criteria**	**0 < Poor <=1**	**1 < Satisfactory <= 2**	**2 < Excellent <=3**
ER6	Code Structure and Design	Poor code structure and missing essential code elements like functions, data structures, or control flow.	Code functions but lacks proper structure, documentation, or uses inefficient algorithms.	Well-structured, documented code using appropriate data structures and algorithms.
#	**Qualities & Criteria**	**0 < Poor <= 1**	**1< Satisfactory <= 2.5**	**2.5< Excellent <= 4**
ER8	Question & Answer	Answers some questions but not confidently or based on lab task knowledge.	Answers most questions confidently and based on lab task knowledge.	Answers all questions confidently and demonstrates a deep understanding of the given lab task.